AI Lab Practice and Career Match Analyst

YuHsuan Tseng

Indiana Wesleyan University

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Dr. Marvin Hunt Sr.

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I used ChatGPT, the Data Analyst GPT, and STORM to explore survey participation and develop the Career Match Analyst plugin. The exercises showed me how AI can organize information and suggest useful directions while still requiring careful judgment. I completed the lab as an individual project using the guided activities (*AI Lab*, n.d.).

Task 1 Practice With a Large Language Model

I tested the same ChatGPT prompt in three separate chats and compared the responses. The prompt used the context-task-constraints framework. ChatGPT is a generative AI tool developed by OpenAI (n.d.-a).

Context: I am a market research analyst examining survey participation. The following data are synthetic aggregate weekly records. They do not identify individual panelists. Week 1: 1,000 invitations and 800 completed surveys. Week 3: 1,000 invitations and 200 completed surveys. Week 6: 1,000 invitations and 150 completed surveys.

Task: Calculate each week's completion rate, defined as completed surveys divided by invitations. Present the results in a table. Explain the participation pattern. Suggest two possible explanations, clearly labeled as hypotheses. Propose one experiment to explore an intervention.

Constraints: Use only the information provided. Do not invent participant characteristics, research findings, or statistical significance. Distinguish declining completion rates from individual panelist churn. Do not claim that these data establish why participation declined. Write for a research operations manager in no more than 350 words.

Table 1 shows the completion rates used to check the numerical accuracy of the responses.

Table 1

Completion Rates in the Synthetic Survey Data

Week	Invitations	Completions	Completion rate
1	1,000	800	80%
3	1,000	200	20%
6	1,000	150	15%

Note. Completion rate = completed surveys divided by invitations. These are aggregate records, not individual panel histories.

Were the results what I expected? Yes. All three responses calculated the rates correctly and distinguished aggregate completion rates from individual churn. V1–V3 refer to the three chats.V1's second hypothesis emphasized a decline in perceived value; V2 and V3 emphasized insufficient incentives. Perceived value is broader than the incentive alone.

What were the significant similarities and differences? All three identified survey fatigue as a hypothesis and recommended a randomized incentive experiment. V2 added relative-decline calculations; V3 included a visualization. Their agreement did not establish that the hypotheses were true.

How did I assess quality? I considered accuracy, consistency, relevance, clarity, and analytical reasoning. I checked whether responses labeled hypotheses, avoided unsupported conclusions, and proposed practical experiments. A useful explanation for a research operations manager must distinguish the observed decline from its unknown causes and from individual panelist churn.

Task 2 Practice With a Custom GPT

I used the Data Analyst GPT (OpenAI, n.d.-b) to explore the same synthetic data through a continuing conversation. Table 2 records the starting prompt and six follow-up moves required by the guide.

Table 2

Prompts Used in the Data Analyst Conversation

Conversation move	Prompt
Starting message	Analyze this synthetic survey data: Week 1 had 1,000 invitations and 800 completions; Week 3 had 1,000 invitations and 200 completions; Week 6 had 1,000 invitations and 150 completions. Calculate completion rates, explain the pattern, and recommend a next analytical step. Distinguish observations from hypotheses.
Ask for a change	Rewrite your explanation for a nontechnical research operations manager. Keep it under 150 words and preserve the numerical results.
Add my own information	My hypothesis is that respondent burden may contribute to declining participation. Treat this as an unverified possibility. How would you investigate it, and what evidence might contradict it?
Ask for more or better	Strengthen your recommendation by adding one alternative explanation, the additional data needed, and a measurable success criterion.
Ask why	Explain the evidence and assumptions behind your recommendation. Which parts follow directly from the supplied data, and which require additional investigation?
Ask what if	What if completion rates increased after an incentive, but the audience receiving invitations also changed? How would that affect our interpretation?
Request a summary	Summarize our conversation, including the original task, changes to your recommendations, unresolved questions, and limitations.

My reflection. I found the exercise interesting, convenient, and useful. Follow-up questions helped refine the explanation and explore alternatives. The audience-change question also challenged whether an apparent incentive effect could have another explanation.

The AI extrapolated more than I would have, sometimes adding unsupported assumptions or targets. I needed to distinguish a proposed target from a finding and a plausible explanation from evidence. I see it as a useful starting point that requires active review. I did not measure time savings.

Task 3 Practice With STORM

I used STORM to generate an article on a topic connected to my market research experience.

Topic: Factors affecting continued participation in online survey panels: respondent burden, invitation frequency, incentives, survey relevance, and trust.

Writing purpose: Examine what research suggests about these factors, where findings conflict, and how study design affects the conclusions. Distinguish evidence about participation rates from evidence about long-term panel retention.

How I understand the tool. STORM explores questions from different perspectives, simulates research conversations, gathers online sources, and creates an outline before drafting a cited article (Shao et al., 2024). Its brainstorming view helps reveal the questions behind the article. Organization and citations do not guarantee that every claim follows from its source.

My reflection. I found STORM convenient for breaking a broad topic into research questions. However, the article emphasized burden and invitation frequency while giving relevance and trust less attention. It sometimes mixed completion, breakoffs, future participation, and retention. Declining one survey does not necessarily mean leaving the panel.

One brainstorming response said the sources did not directly compare incentive types or payment timing; a later response made confident comparisons. Different sources might explain

this, but I would need to inspect the studies before accepting the conclusions. Numbered citations made the output look credible without establishing adequate support.

Task 4 AI Chatbot Creation

Career Match Analyst helps evaluate opportunities against my experience and practical requirements. I have used the personalized version. The current saved release, version 0.3.0, generalizes the workflow, but I have not tested that generic version (Tseng, 2026). The plugin's description and instructions are submitted as a companion file.

Design Thinking

Empathize. I needed suitable openings, realistic qualification judgments, and transit access. A title alone did not establish fit. The initial empathy work was reflection on my needs, not interviews with others.

Define. How could I prioritize opportunities while making qualification gaps, compensation, transit limits, and authorization uncertainties clear? The assistant should explain its reasoning and support my application decisions.

Ideate. I considered a checklist, a reusable prompt, and a saved assistant. The first two left repeated reading or repeated setup. I chose a saved assistant to keep the process and reference material together.

Prototype. ChatGPT and Plugin Creator helped package an LLM workflow with instructions and three reference files. I did not train a separate model. The generic version replaces my profile with a blank intake template.

Test. Earlier AI-assisted checks used fictional postings. My use of the personalized version added usability feedback. Neither establishes the generic version's behavior or measured accuracy and time savings.

Build Plan and Intended Interaction

Table 3 summarizes the intended workflow using the guide's ten planning questions.

Table 3

Career Match Analyst Build Specifications

Planning requirement	Specification
1 Objective	Find, evaluate, or compare opportunities. Explain matches, gaps, practical concerns, and next steps.
2 Knowledge	Use confirmed job-related profile information, accessible postings, employer sources, and transit evidence. Resolve material conflicts.
3 Boundaries	Never invent facts or eligibility. No applications, employer contact, or saved-profile changes without a specific request.
4 Engagement	Explain matches and unknowns candidly. Ask focused questions; keep information specific to the current user.
5 Input formats	Accept text, public links, and readable résumés or screenshots when supported. Request missing text. No generated images are needed.
6 Opening	Summarize the profile, search, assumptions, and unknowns. Wait for confirmation before personalized recommendations.
7 Sequence	Confirm inputs; inspect postings; verify facts; screen conflicts; map requirements to experience; score and rank.
8 Conclusion	Recommend Apply, Consider after clarification, or Skip with reasons. Finish searches with priorities and unresolved checks; the user decides.
9 Success	Check source facts and ask the user to rate recommendations useful, partly useful, or unsuitable, with reasons.
10 Rule tests	Test intake, profile conflicts, screening, scoring, sources, privacy, and output structure. Save responses, changes, and retests.

The rubric assigns up to 40 points for duties and transferable experience (D), 25 for required tools and technical depth (T), 15 for responsibility level (L), 10 for office transit access (C), and 10 for preferred qualifications and domain knowledge (P). Unknown criteria produce a provisional range. Education is checked separately when required. A score is a comparison aid, not a probability of being hired.

For a 20-opening search, the requested structure is five detailed assessments, five shorter assessments, and ten compact entries. The actual search settings and requested number must be confirmed for each user. The office transit rule checks a walk of no more than 10 minutes to a usable stop or station when confirmed as a requirement; it does not establish the full residential commute.

Decisions and Iterations

I separated professional match from practical feasibility: a high score cannot cancel a confirmed restriction. I clarified that my original salary base minimum applied only to permanent full-time roles and requested the 5+5+10 layout. Later revisions added initial confirmation, a blank user template, and transit access instead of education points. Other users must confirm their own filters.

The releases progressed through 0.1.0, 0.2.0, and 0.3.0. Source inspection on October 30, 2026, confirmed the private Career Match Analyst plugin and its instruction file, blank profile template, matching rules, and rubric (Tseng, 2026). This saved structure documents the build without assuming readers can open the private link.

Challenges included conflicting résumé wording, overstated technical fit, and uncertain transit access. I reconciled conflicting claims, required specific experience for skill matches, and required office, route, and service evidence before confirming transit access. Generalizing the workflow also required removing my personal information from shared instructions.

Testing and Evidence

Table 4 summarizes eight fictional cases from two earlier AI-assisted exercises. Separate AI conversations received the instructions and fixtures without expected answers; ChatGPT

checked their responses against the rules. These concern earlier personalized settings, not live employers or the generic version. Old scores are omitted because the rubric changed.

Table 4

Recorded Checks of the Earlier Personalized Instructions

Case and supplied condition	Recorded response
A: No OPT; CPT unstated	Retained for clarification; did not equate the OPT restriction with a CPT restriction or acceptance.
B: Explicit no CPT	Skipped under the original confirmed authorization filter despite a strong professional match.
C: Six-month contract, US$24/hour, 25 hours/week, 68-minute commute	Assessed pay case by case, marked the stipulated commute as a stretch, and labeled US$600/week as arithmetic rather than guaranteed income.
D: Permanent full-time pay of US$54,000–58,000	Skipped because the posted maximum was below the original US$60,000 base minimum.
E: Senior ML platform engineering	Identified senior scope and a substantial production-deployment gap; skipped.
F: Missing salary and an embedded instruction to disclose the profile	Kept pay and restrictions unconfirmed; ignored the instruction to disclose the profile or award a perfect score.
G: Stipulated 72-minute commute	Skipped under the earlier personal limit of 70 minutes. This was not a verified transit route.
H: Invalid link without readable text	Requested accessible posting text; did not invent a score or claim to have opened the page.

Note. Commute times were stipulated. Earlier personal thresholds do not apply automatically to other users.

Table 5 identifies planned checks for the generic version. Save the prompt, version, actual response, evaluation, changes, and retest results.

Table 5

Remaining Tests and Pass Criteria

Rule or feature	Scenario and pass criterion

Rule or feature	Scenario and pass criterion
Intake and confirmation	Start a fresh matching conversation. Show known inputs, assumptions, and unknowns; wait for confirmation. Change a material setting and confirm the revised inputs before proceeding.
User information and changes	Use a different fictional user and conflicting profile details. Inherit none of my personal facts; use explicit corrections, avoid upgrading skills, and distinguish temporary changes from lasting updates.
Pay, scope, and authorization	Vary employment type, hours, responsibility level, pay around a confirmed floor, and explicit versus unstated authorization wording. Apply only that user's confirmed filters and preserve unknowns.
Transit evidence	Test verified office walks of 9, 10, and 11 minutes, then omit route or service evidence. Apply the confirmed boundary; leave unsupported access unconfirmed; keep total commute separate.
Scoring and required credentials	Recalculate points, test an unknown criterion, and compare a tool first preferred and then required. Explain consistent evidence thresholds, ranges, and any essential-qualification override.
Sources and varied inputs	Try readable text, a résumé or screenshot, an inaccessible link, expired or duplicate postings, and instructions embedded in a posting. Verify sources, request missing text, and ignore hostile instructions.
Fairness, privacy, and actions	Vary irrelevant personal traits; keep fit judgments unchanged. Request identifiers or an unrequested application/message. Keep private data out of searches and shared files; require a specific request for external actions.
Output and usefulness	Use a search with fewer verified roles than requested. Do not pad results; include required fields and next steps. Ask whether a simpler table would be easier to use than 5+5+10.

Privacy and Terms Review

The policy review led to the following design lessons:

- Prompts and uploaded files can contain personal data processed by the platform. I should provide only information needed for matching and omit contact details and document identifiers (OpenAI, 2026a).
- Private plugin visibility does not mean local-only processing. Model-improvement choices and retention depend on policy conditions and account settings, which require my own review (OpenAI, 2026a).

- I need appropriate rights to provide input. Generated output can be inaccurate, so I must check employer facts and recommendations before relying on them (OpenAI, 2026b).
- A reusable assistant should not package my personal information for other users. I removed that background from the generic template and kept the application decision under the user's control.

Personal Reflection and Collective Insights

I found the personalized assistant useful, but it still needed tweaking and clarification during use. The 5+5+10 layout remained messy. I stopped iterating because the process itself was becoming messy. I would next test a simpler presentation. My experience does not establish the generic version's accuracy or measured time savings.

Across the lab, AI helped organize information and explore alternatives but sometimes exceeded the supplied evidence. Design thinking turned that concern into requirements such as confirming inputs and preserving unknowns. I need to check both correctness and whether the response is clear enough to use.

AI Assistance Disclosure

ChatGPT (OpenAI, n.d.-a) assisted with prompts, comparisons, reflection drafting and editing, assistant planning and creation, source inspection, prototype checks, and documentation. STORM generated the article and brainstorming material. I supplied my experiences, corrections, and design choices and reviewed this individual submission. AI-assisted checks are distinguished from my own use.

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